

F24-057-D-ARch360

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Contents

1	Introduction	1
1.1	Problem Statement	3
1.2	Scope	3
1.3	Modules	4
1.3.1	Module 1: Building Instantiation	4
1.3.2	Module 2: Area Analysis	4
1.3.3	Module 3: Customization/Decor	4
1.3.4	Module 4: Building Tour	5
1.3.5	Module 5: AI Expert Module	5
1.3.6	Module 6: UI Module	5
1.4	User Classes and Characteristics	6
2	Project Requirements	7
2.1	Use-case	7
2.2	Functional Requirements	10
2.2.1	Building Instantiation Module	10
2.2.2	Area Analysis Module	10
2.2.3	Customization/Decoration Module	10
2.2.4	Building Tour Module	11
2.2.5	AI Expert Module	11
2.3	Non-Functional Requirements	11
2.3.1	Reliability	12
2.3.2	Usability	12
2.3.3	Performance	12
2.3.4	Security	12
2.4	Domain Model	13
3	System Overview	14
3.1	Architectural Design	15
3.2	Design Models	18
3.3	Data Design	22

List of Figures

2.1	Use Case Diagram	9
2.2	Domain Model	13
3.1	Architecture Diagram	17
3.2	Activity Diagram	18
3.3	Class Diagram	19
3.4	ERD	20
3.5	Class Level Sequence Diagram	21

List of Tables

1.1 User Classes and Descriptions 6

Chapter 1

Introduction

This document aims to officially present ARch360, a new augmented reality solution aimed at complementing the architectural design process. This document describes ARch360, its motivation, system design, and goals to facilitate communication between clients and architects via immersive visualizations in 3D for construction projects.

The proposal goes on to illustrate how the application uses AR technology, powered by Unity and AI, to allow clients an account of decisions before construction even begins with a promise for more precise project outcomes. This means taking conventional architect drawings and turning them into 1:1 scale interactive 3-D holograms. A mobile device enables clients to navigate through a virtual representation of their future homes or commercial spaces. This approach enables clients to:

- **Interact with their space:** Allows changing wall colors and tile designs of their future homes.
- **Reduce uncertainty:** By using augmented reality (AR) to experience their future environment in real-time, clients can clearly understand the design choices that will minimize last-minute modifications.

System Design:

- **Architecture Designs:** Starts with standard AutoCAD or architectural files, converted into 3D models.
- **Unity Integration:** The 3D models are integrated into Unity, which handles the visualization and AR components.
- **Augmented Reality (AR) Visualization:** Using mobile applications Clients can experience these models in real-time AR, viewing their spaces in their actual size and scale.

- **AI-Assistance:** The project integrates AI to enhance design suggestions based on user preferences, hence optimizing the user experience.

Intended Audience for the Arch360 Document

- **Clients (Homeowners, Business Owners, Property Developers):**
 - **Role:** To make sure that the final product meets their expectations, end users must be able to visualize and interact with architectural ideas.
 - **Interest:** Better understanding of their future spaces and environments, less uncertainty, and increased satisfaction with the final design.
- **Architects:**
 - **Role:** Before construction begins architects, can communicate architectural visions to their clients through ARch360.
 - **Interest:** This applies to enhancing the design process and the reduction of misunderstandings between employees, clients, and other stakeholders to improve on organizational communication and customer satisfaction.
- **Interior Designers:**
 - **Role:** People who have to choose the colors, the materials, and other practical and aesthetic features of the interior of the space.
 - **Interest:** Presenting possible designs to the clients and making changes on the work in response to the opinion of the clients.
- **Construction Contractors:**
 - **Role:** The workers that have the responsibility of translating the concept into a tangible construction.
 - **Interest:** The ability to become a clear and accurate receiver of design instructions such as avoiding situations where construction work has already progressed only for client to make changes that will require other parts of the construction project to be done again.
- **Project Managers:**
 - **Role:** People in charge of coordinating finances, schedules, and coordination between contractors, architects, and designers during the whole construction process.
 - **Interest:** Ensuring that the project runs efficiently, minimizing revisions, and adhering to the client's vision while staying on schedule and within budget.

- **Investors or Real Estate Developers:**
 - **Role:** People who provide resources and finances for construction.
 - **Interest:** Reducing the risk of cost overruns and delays by ensuring design clarity and client approval early in the process.
- **AR and Unity Developers:**
 - **Role:** The entire team responsible for developing ARch360 project.
 - **Interest:** Continuous development, system improvements, and ensuring user satisfaction with the platform's performance.
- **Supervisors and Academic Advisors:**
 - **Role:** Mentors who guide the development of the ARch360 project from a research and technical perspective, ensuring the solution is academically sound and meets project goals.
 - **Interest:** Ensuring that the project is innovative, and practical, and delivers a real-world solution to the identified problem.

1.1 Problem Statement

Provide a problem statement in a concise paragraph describing: Why you are developing this system? What problem does your software solve? (Usually in 10-16 sentences)

1.2 Scope

The architecture and construction industries often face a significant challenge in bridging the communication gap between clients and professionals. Clients typically struggle to visualize 2D plans or even traditional 3D renders, resulting in misunderstandings, dissatisfaction, and costly revisions during or after the construction phase. These miscommunications arise because clients are unable to fully grasp the scale, design details, or spatial layout of their future spaces through static visuals. Furthermore, last-minute changes due to unmet expectations can lead to project delays and budget overruns, impacting the overall efficiency of the design and construction process.

To address this issue, we are currently creating **ARch360**, an application that will apply the use of Augmented Reality to present architectural designs in the form of interactive 3D environments. Through this system, clients can get a feel of their future homes or commercial space before construction takes place. Through real-time walkthroughs, and

the ability to interact with the design and view the changes instantaneously, **ARch360** minimizes potential misunderstandings and guarantees the result to be satisfactory for the client. It will increase clients' engagement with the architects, decrease the likelihood of the change orders, and improve clients' satisfaction by providing a more effective and enjoyable way to design and modify spaces. In the end, with ARch360, the design cycle becomes less cumbersome, and time could be saved while also eliminating the chances of its unsatisfactory completion.

1.3 Modules

1.3.1 Module 1: Building Instantiation

1. **Model Loader:** Loads 3D architecture models into Unity (such as AutoCAD files) for augmented reality (AR) rendering. This enables users to see how their building might look in the actual world.
2. **Placement Tools:** Users can use the tools they need to precisely position and align virtual architectural models using augmented reality in real space. This guarantees that the 3D model blends in with the surroundings.
3. **Scale and Alignment:** Automatically adjusts the scale and alignment of the 3D models to match the real-world dimensions, ensuring that the model's size and position are proportionate and realistic.

1.3.2 Module 2: Area Analysis

1. **Environment Scanning:** Utilizes ARCore to scan the user's physical environment, capturing detailed spatial data such as walls, floors, and obstacles.
2. **Spatial Mapping:** Creates a comprehensive 3D map of the scanned area, which is used to integrate the virtual building model seamlessly with the actual environment.
3. **Measurement Tools:** Offers accurate tools that enable users to measure distances and areas in the scanned space so they have a strong idea of how best to fill out spaces once design decisions are finalized.

1.3.3 Module 3: Customization/Decor

1. **Wall Customization:** Users can see how different colors look on their walls by customizing the walls in their 3D building model by choosing various color schemes.

2. **Floor Customization:** Allows the user to choose the most appropriate tiling options available according to their preferences and the interior of their homes or spaces. Tiles can be placed on the floor, and users can easily visualize and have an idea which tile options best match the interior of their homes.

1.3.4 Module 4: Building Tour

1. **Virtual Tours:** A First person controller is implemented to allow users a virtual tour of the building and navigate different floors and rooms for a better understanding of the design.
2. **Floor Navigation:** Provides easy navigation tools for moving between different levels of the building, enhancing the immersive experience.

1.3.5 Module 5: AI Expert Module

1. **Design Suggestion Engine:** The AI Expert module in the ARchitect360 application utilizes open-source tools and frameworks to provide intelligent design suggestions, helping users create personalized and stylish architectural designs. By leveraging pre-trained language models, this module offers tailored recommendations for design elements such as wall colors and tile schemes based on user preferences and architectural styles.

1.3.6 Module 6: UI Module

The **UI Module** is responsible for all user interface elements and interaction methods, ensuring seamless navigation and intuitive control within the Android mobile app. This module focuses on delivering a fluid and responsive user experience tailored for a mobile environment, and will be designed using **Figma** for visual layouts, with Unity as the development platform.

1. **Touch Control:** Provides intuitive touch gestures for users to interact with the app. Includes tap, swipe, pinch-to-zoom, drag, and long-press actions for selecting, placing, and customizing elements within the 3D environment.
2. **Gesture-Based Navigation:** Implements mobile navigation gestures, such as swiping between screens, double-tap for zooming in/out, and drag for rotating or moving elements in the 3D space.
3. **Main Menu Screen:** The central hub for accessing core functionalities, including starting a new project and loading existing projects.

4. Customization Panels: Mobile-friendly panels for wall and floor customization. Users can select wall paint colors and tile designs through swatches and apply them in real-time on their 3D models. Includes real-time preview functionality.
5. Building Tour Controls: Virtual joy stick or swipe-based controls for navigating through the virtual building and a simple touch-based floor selection options for exploring different levels of the building design and options for first-person to enhance the immersive experience.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

Table 1.1: User Classes and Descriptions

User class	Description
Customers	Customers use the mobile app to visualize their future homes or commercial spaces in AR. They can customize design elements (e.g., wall colors, floors) using intuitive touch controls and check the alignment of the final design with their expectations.
Architects	Architects utilize ARch360 to showcase designs to clients, make on-the-spot modifications, and promote transparent communication. They work across desktop computers for design creation and mobile devices for augmented reality presentations, staying actively engaged with clients throughout the design process.
Interior Designers	Interior designers concentrate on visualizing and personalizing interior features such as color schemes and décor. They utilize AR-based customization tools for instant previews, engaging primarily during the interior design stage, with most interactions taking place through mobile devices.
Project Manager	Project managers are responsible for monitoring the project's progress, ensuring it remains on schedule and within the allocated budget. They rely on ARch360 for comprehensive overviews, client input, and tracking timelines, frequently engaging through mobile.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

1. UCO1:Usecase: View Building in AR

- **Actor:** Client
- **Type:** Primary
- **Main Flow:** The AR view is started by the client.Using the device camera, the system superimposes the 3D model on top of real-world environment.

2. UCO2:Usecase:Import 3D Model

- **Actor:** Client
- **Type:** Primary
- **Main Flow:** Client clicks import button to import a 3D model into the repository.Client chooses an appropriate file type of .fbx format. The model is loaded and shown.

3. 3. UCO3:Usecase:Analyze Area

- **Actor:** Client
- **Type:** Secondary
- **Main Flow:** The client select area analyze tool.The client defines the point from which the system start measuring. The client is able to see the results after area analyses.

4. **UCO4:Usecase: Provide a 3D model.**

- **Actor:** Architect
- **Type:** Primary
- **Main Flow:** Architect creates and sends 3D model to the client of .FBX file format. The client receives and imports the model.

5. **UCO5:Usecase: Customize Walls**

- **Actor:** Client
- **Type:** Secondary
- **Main Flow:** The client chooses the wall customization choice.The device provides several wall design options.The client implements the desired customization.

6. **UCO6:Usecase: Recieve Design Suggestions**

- **Actor:** Client
- **Type:** Secondary
- **Main Flow:** The client clicks on the receive generate suggestions option. Client is shown different suggestions related to the customization he has made to the 3D model. The client uses most appropriate design suggestions and make changes according to that.

7. **UCO7:Usecase:Navigate Building**

- **Actor:** Client
- **Type:** Primary
- **Main Flow:** The client clicks on navigate building option. Client is given the option to select which floor he wants to navigate. When client selects floor client can select options to navigate different rooms in the building.

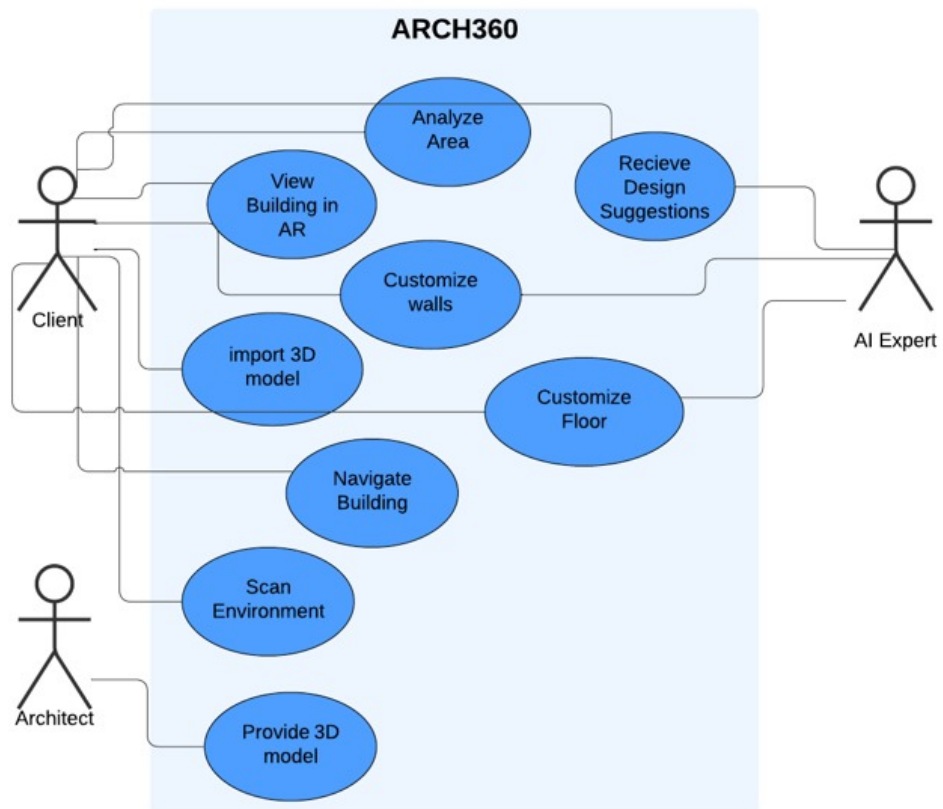


Figure 2.1: Use Case Diagram

2.2 Functional Requirements

The **ARch360** system should provide an immersive augmented reality experience for visualizing and customizing architectural designs. These requirements are functional in nature, divided into modules, and define the interface each module should expose to others.

2.2.1 Building Instantiation Module

This module shall load and render 3D architectural models in an AR environment.

- **FR1:** The system should be able to import 3D models in various formats such as .FBX, .OBJ, or cad files of AutoCAD into the AR environment.
- **FR2:** It should have utilities to orient and place the virtual building models in the real space with much accuracy.
- **FR3:** This system shall have the ability to automatically scale and align the 3D models into real-world dimensions.

2.2.2 Area Analysis Module

This would be responsible for scanning and mapping the user's physical environment to ensure seamless integration of the virtual model.

- **FR1:** The system shall make use of the ARCore in scanning the environment of the user, capturing the spatial data like walls, floors, obstacles.
- **FR2:** The system shall build a complete 3D map of the area that gets scanned to integrate the virtual building model seamlessly.
- **FR3:** The system shall be able to provide functionalities to measure distances and areas in the scanned environment.

2.2.3 Customization/Decoration Module

This module allows users to personalize visual features of their building designs.

- **FR1:** The system shall be able to allow users to select different colors for the walls in their 3D building model.

- **FR2:** The system shall be able to enable the user to select from a set of tile designs and styles for the floors.
- **FR3:** The system shall show, in real time, the selected customizations on the 3D model.

2.2.4 Building Tour Module

This module provides the interactive exploration of the virtual building.

- **FR1:** The system shall implement an FPS controller in order for users to move inside the virtual building.
- **FR2:** The system shall provide tools for navigation among the floors and rooms.
- **FR3:** The system shall embed the overview map to help users travel around large or complex buildings.

2.2.5 AI Expert Module

The module is responsible for providing intelligent design suggestions to the user.

- **FR1:** The system shall make use of open-source tools and frameworks that shall give AI-driven design suggestions to the end user for things such as wall colors and tile schemes.
- **FR2:** The system shall employ the usage of pre-trained language models to form recommendations based on user preferences and architectural style.
- **FR3:** The system shall enable a user to accept, modify, or reject design suggestions proposed by AI-driven architecture.

These functional requirements ensure **ARch360** will deliver a complete-scale and user-friendly AR experience for architectural visualization and customization.

2.3 Non-Functional Requirements

Here are the Non-Functional Requirements (NFRs) for ARch360:

2.3.1 Reliability

- **Uptime:** ARch360 should have an uptime of 99.9%, ensuring high availability.
- **Error Recovery:** In case of failure, the system should automatically recover from crashes within 10 seconds, saving user progress where applicable.
- **Data Integrity:** All data (e.g., project designs, user settings) must be saved reliably, with no data loss in case of system crashes.

2.3.2 Usability

- **Ease of learning:** A system should be simple enough so that users can understand how to perform simple tasks easily.
- **Error Avoidance and Recovery:** ARch360 should be able to resolve user errors with a clear and user-friendly interface.
- **Accessibility:** ARch360 to be available and operational 24/7 ensuring it is available when users need it.
- **USE-1:** The system shall allow users to upload a CAD file and visualize it in AR within two interactions.

2.3.3 Performance

- **PER-1:** The system shall load AR models within 7-8 seconds on mobile devices and within 5 seconds on desktop devices.
- **Rendering:** ARch360 must maintain a frame rate of at least 30 FPS on mobile devices and 30 FPS on desktops for smooth interaction with 3D models.

2.3.4 Security

- **Authentication:** All users must authenticate via secure login, with multi-factor authentication (MFA) for high-privilege users (e.g., architects, developers).
- **Data Encryption:** All user data, including design files, must be encrypted in transit (using TLS) and at rest (AES-256).
- **SEC-1:** The system should resist unauthorized access and maintain the log of all login attempts.

2.4 Domain Model

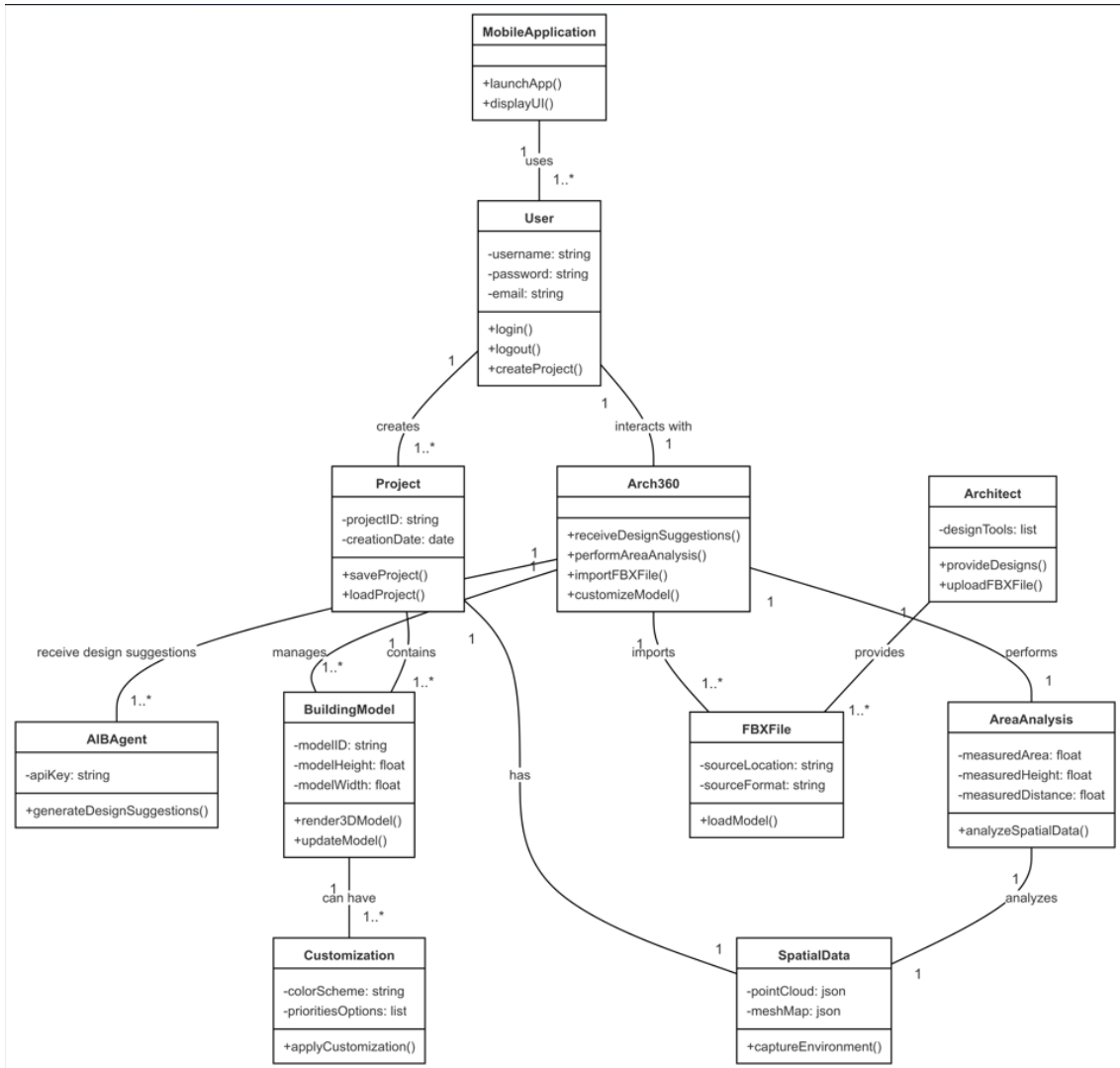


Figure 2.2: Domain Model

Chapter 3

System Overview

The ARchitect360 Project, through the use of extended reality-based tools, aspires and seeks to revolutionize both the creation and presentation of architectural designs. Distinctively, it enables a person to engage with three-dimensional structures within a physical space long before a single brick has been laid, thereby providing potential clients, architects, and contractors with an engaging and collaborative tool during the period of building design. Most specifically, the platform is intended to handle general problems in design communication, including misinterpretation of blueprints and the inability to visualize spaces realistically. ARch360 is a complex transformation of 2D architectural drawings into dynamic 3D models that clearly show how one can understand, explore, and make changes in building designs

1. **Capability:** The platform is integrated with five major modules, which include:
2. **Building Instantiation:** This module instantiates and renders 3-D models such as Auto-CAD files in Unity for AR visualization. It allows users to position the model correctly within their real-world space and provides utility for tweaking the scale while manually aligning it to ensure the building best fits in a real-world space.
3. **Area Analysis:** By utilizing the ARCore technology, this module scans the user's surroundings to record the spatial data, thus creating an environmental map in 3D. It provides tools to measure distances and assess areas; those are important in evaluating the feasibility of the designs against particular sites.
4. **Customization/Decor:** The purpose of this feature is to let the user customize the graphical aspects of their building designs. It allows the user to make aesthetic changes, such as wall colors and floor designs, and to see what those changes would look like in real-time. Building Tour: The platform integrates an FPS controller that lets users tour the virtual building in a very interactive way. They can go up

and down floors, rooms, and areas to create full insight into the layout and spatial relations.

5. **Building Tour:** The platform integrates an FPS controller that lets users tour the virtual building in a very interactive way. They can go up and down floors, rooms, and areas to create full insight into the layout and spatial relations.
6. **AI Expert Module:** This module uses AI-powered suggestions for design elements like wall colors and flooring options. Pre-trained language models use AI to make personalized suggestions based on user preferences and architectural style.

Context and Background: Traditionally, architectural visualization has been done using 2D blueprints, renderings, and physical models, resulting in communication barriers between architects and their clients. Many times, there were errors in the understanding of how the final images of their places would look and feel because these were beyond the comprehension of the non-technical clients. This resulted in the need to make extremely expensive changes during the construction process.

The project exploits new technologies and open-source frameworks, including Unity and ARCore tools and AI models, granting thus rapidity in development, more opportunities, and wider access.

Design: The ARch360 is designed with great tenderness to the users allowing technical end users a complicated user interface. Practical application for integration of the 3D models happens quite effortlessly into the physical environment. Evaluation of design solutions is provided much closer to reality. Due to Rapid alterations of styles, thanks to the application of Artificial Intelligence, design elements can be tried out and the effects of such decisions witnessed instantly. It is a Building Designer type of software in which all the modules are independent and each module takes up a certain part of the architectural visualization process and also allows easy modifications or additions for different projects on the system

3.1 Architectural Design

The initial concept behind ARchitect360 was to create architecture that rests on a modularly scalable structure, wherein each component is independent yet interacting with other components to perform flawlessly and assure smooth performance. The perspective view of multiline architecture, where the core functionalities are separated into distinct layers, goes to include: UI, business logic, and data management. This modular structure allows for easier development, maintenance, and future extension of the system.

At the highest level, the system is decomposed into five core modules responsible for

different aspects of the architectural visualization process: Building Instantiation, Area Analysis, Customization/Decor, Building Tour, and the AI Expert Module. Interaction via a central communication bus among the above-mentioned modules ensures smooth data flow and workflow among them. These are the relationships and, where applicable, interactions across these modules:

1. **Building Instantiation:** This module brings in 3D models into an AR environment. It provides placement tools to align and scale the model correctly based on the user's physical space. It communicates with the Area Analysis module to receive environmental data so that the virtual model is well-matched and fits within real-world surroundings.
2. **Space Analyzer:** ARCore does space scanning and then spatial mapping. This module feeds the system with details of the size of, and space constraints in, the physical environment. It interacts with the Building Instantiation module to position and scale the 3D model appropriately for precise placement within the physical space.
3. **Customization/Decor:** The Customization/Decor module will let the user personalize certain parts of their design, including wall colors and the type of floor. It fetches 3D model data from the Building Instantiation module and refreshes the model with user input. Besides that, it also communicates runtime visual feedback back to the UI layer.
4. **Building Tour:** Building Tour module implements an FPS controller allowing the user to virtually navigate around a building. It works in collaboration with the Building Instantiation and Customization/Decor modules to ensure that navigation of the user within the 3D model is at its full customized state.
5. **AI Expert Module:** This module provides intelligent design suggestions through the use of pre-trained AI models.. Suggests design elements, such as wall colours and tile styles, based on user preference or project requirements. It also pulls in user preferences and building data from the Customization/Decor module to personalize its recommendations and feedback for the user.

High-Level Module Interaction At a high level, these are divided into three layers:

Presentation Layer (UI): handles user input, visual output; AR visualization, and customization interface.

Business Logic Layer: Comprises modules such as Building Instantiation, Area Analysis, AI Expert Module, etc., which are employed for the necessary core processing functions.

Data Layer: It concerns the maintenance of 3D model data, scan data of an environment, and customization parameters.

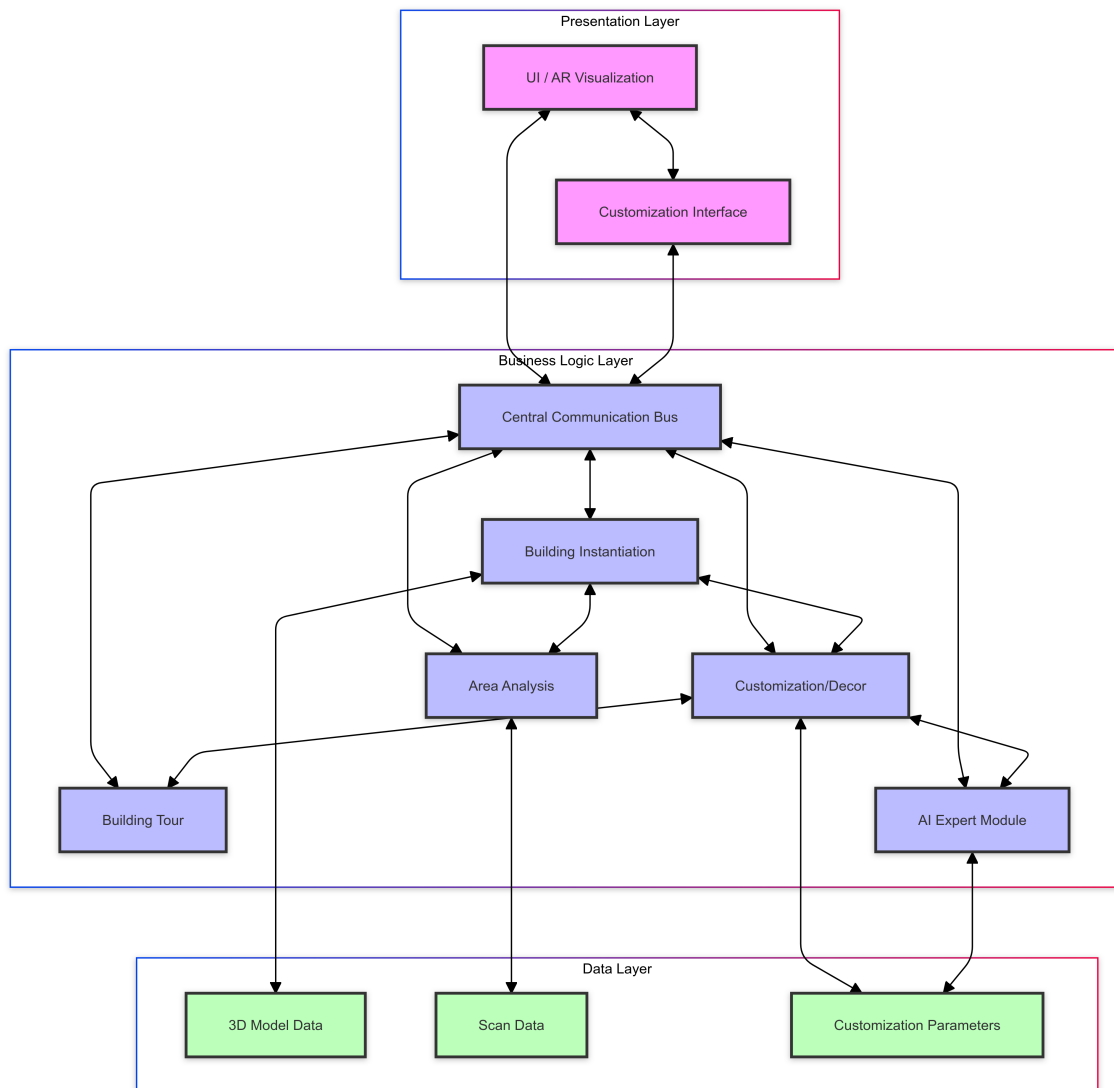


Figure 3.1: Architecture Diagram

3.2 Design Models

Design Models for Object Oriented Development Approach

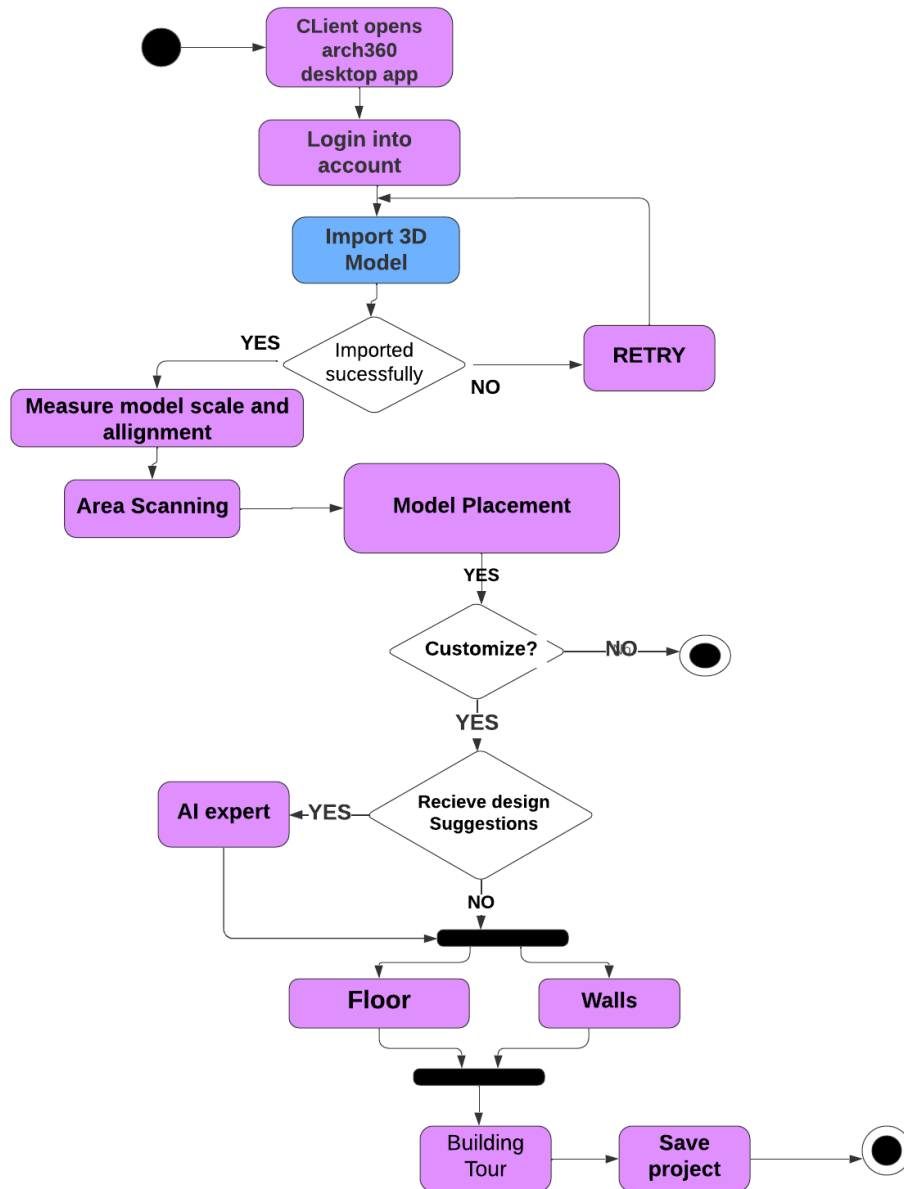


Figure 3.2: Activity Diagram

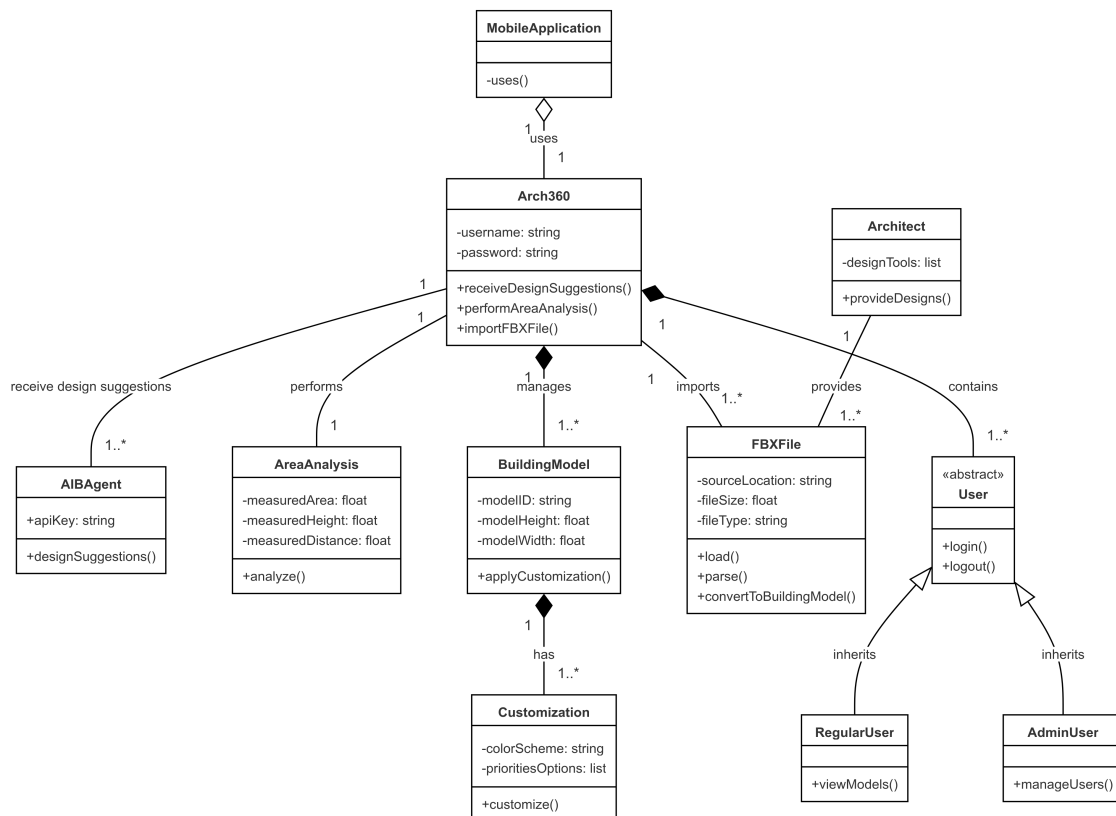


Figure 3.3: Class Diagram

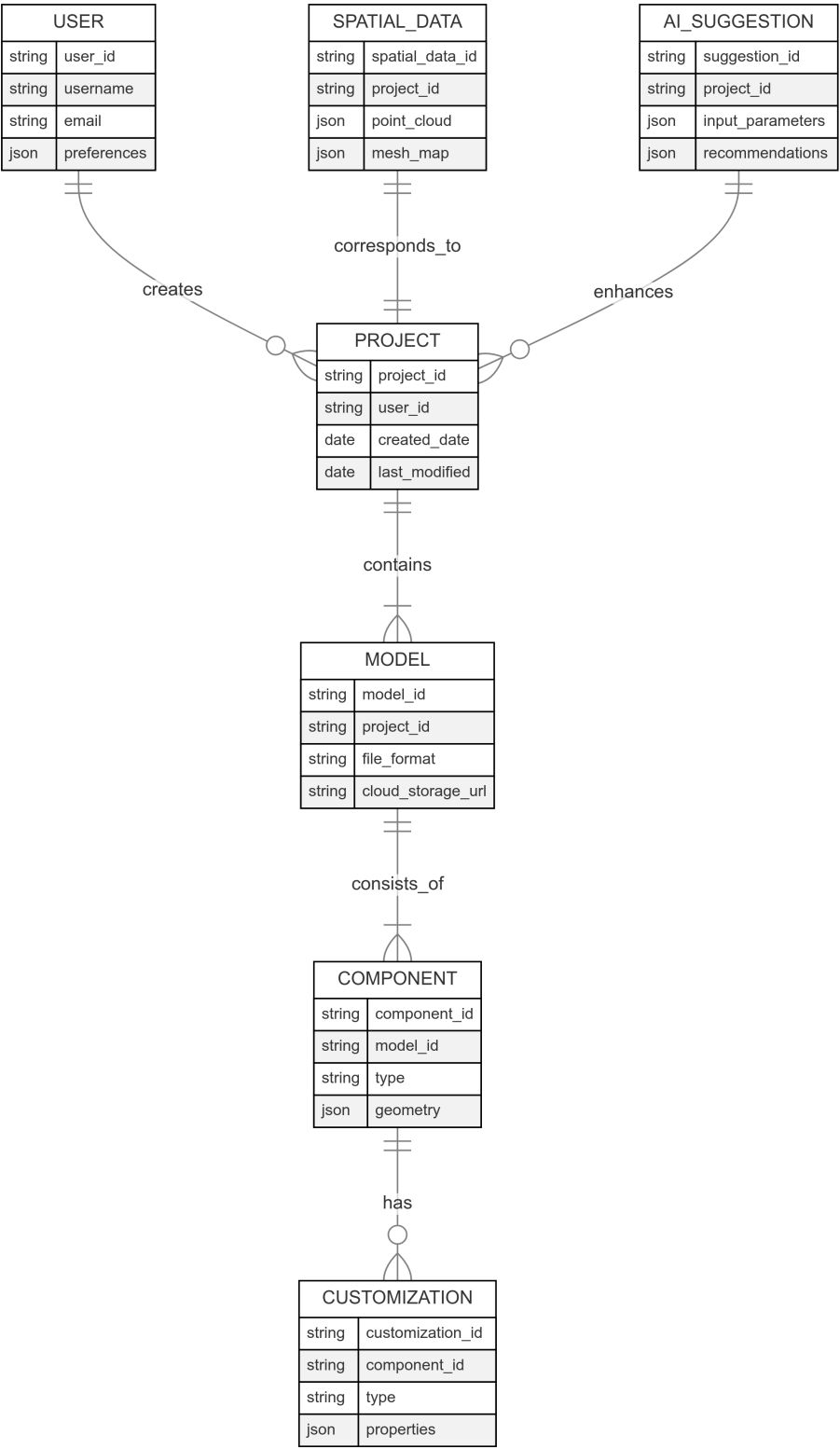


Figure 3.4: ERD

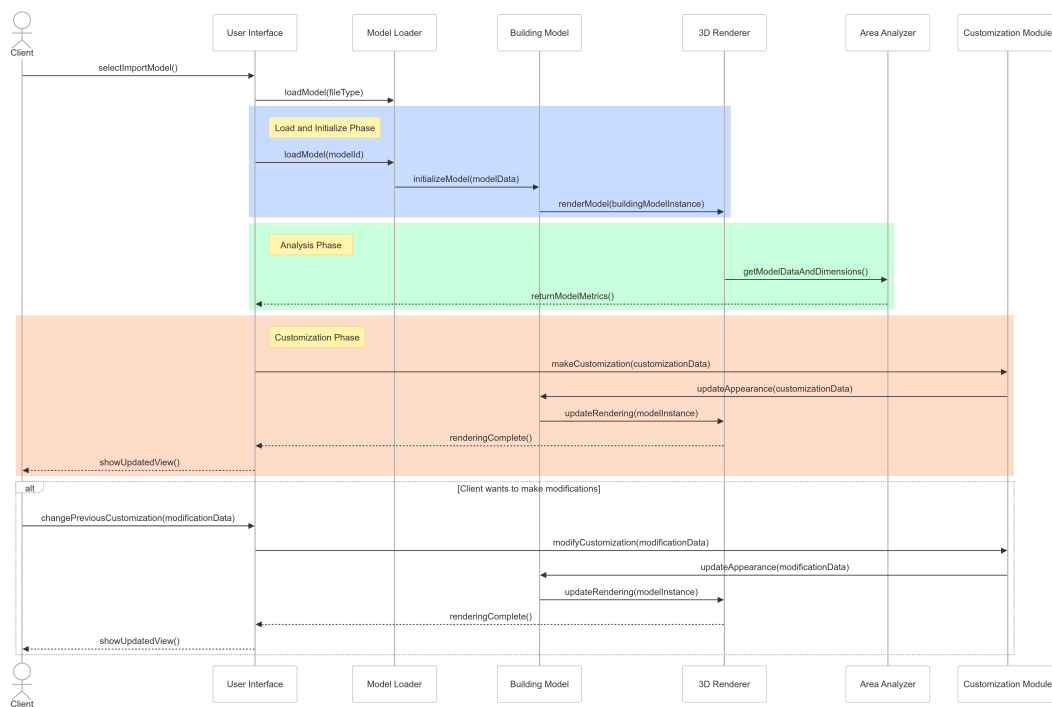


Figure 3.5: Class Level Sequence Diagram

3.3 Data Design

Design Models for Procedural Approach In ARchitect360, the information domain of the system is transformed into a line of data structures that regulate architectural models, user preferences, spatial data, and AI suggestions of the system. These data structures then enable the system to provide complex functionality, including rendering 3-D models in AR, customization of design, scanning the environment, and AI-driven design suggestions.

Data Structures and Entities

1. Architectural 3D Models

- **Data Structure:** The 3D models are stored in any format like .FBX, .OBJ, or AutoCAD. Unity parses the models and renders them by using mesh structures comprising vertices, edges, and faces that define the geometry of the building.
- **Storage and Processing:** The geometries and models are loaded into Unity's MeshRenderer, which processes them for visualization. Models can be stored locally or fetched from cloud storage depending on the system configuration.
- **Organization:** In Unity, it is organized by different building components like walls, floors, and ceilings, structured into a scene hierarchy. This structuring would permit users to interact with and make their own adjustments to individual elements.

2. User Customization Data

- **Data Structure:** Wall colors, flooring, and decor customization options are kept in the form of either key-value pairs or as objects in JSON. Every option is tagged with model parts the respective option belongs to, for example, wall, floor, etc.
- **Storage and Processing:** When a user chooses an option to customize, the system will pick up an appropriate file for a certain material or texture involved- such as wall colors or floor patterns- and apply it to the relevant component of the given 3D model. This is handled dynamically in Unity using material shaders.
- **Organization:** It stores information about user and session personalization; hence the system will be able to remember user preferences across sessions by storing them in a NoSQL database like MongoDB or Firebase.

3. Environment Scanning Data (Spatial Data)

- **Data Structure:** The spatial data originates from environment scanning through ARCore. This information is stored as point clouds and mesh maps. Such point clouds contain 3D coordinates that represent a user's environment geometry.
- **Storage and Processing:** ARCore processes the captured spatial data in real time by creating a spatial map representative of the particular physical environment. The data is to be used by the module of Area Analysis for accurate placing of the virtual 3D models into the real world.
- **Organization:** Spatial data is organized in a 3D grid or octree structure so that efficient mapping of large environments is possible. Each node in the grid represents a real-world location for accurate positioning and placement of virtual objects.

4. AI Data Structure Design Suggestions:

- **Data Structure:** Pre-trained language models and decision trees make the AI-driven suggestions. The input parameters of the AI model are text or numeric input data, such as user preferences (color, lighting, furniture); room dimensions; and style preferences. A set of recommended design options shall be the output based on the input.
- **Storage and Processing:** Storage and Processing shall be done by the AI, which shall process the user's input and give certain design recommendations. These are stored in JSON format and utilized through customization options applied to the 3-D models.
- **Organization:** The AI data generated shall be stored on a per-user profile basis, with suggestions connected to specific portions of the architectural model for personalized changing recommendations.

5. Data Structure: User Profile and Session Data:

- **The architectural design of the system** will decide whether or not the user profiles reside in a relational database, such as PostgreSQL, or otherwise in the NoSQL database, MongoDB. With the profile, user information includes past projects, preference about customizations, and AI-generated recommendations.
- **Storage and Processing:** User profiles will be persisted within the system so that users can revisit their previous choices in design, view recommendations, and make changes between sessions.
- **Data Organization:** The data from the User is stored in tables or documents. For each given user, there is an ID that connects, with that ID, their prefer-

ences, session data, and chosen architectural models within large parts of the system.

Databases and Data Storage ARchitect360 uses both Relational and NoSQL databases for managing the information inside the system:

- **NoSQL Database**-e.g., Firebase or MongoDB
 - **Used for:** User customization data, spatial maps, and real-time session information
 - **Why:** NoSQL databases are best suited for dynamic and flexible data, such as user preferences, which may vary from user to user and might change over time.
- **Relational Database**-e.g., PostgreSQL
 - **Used for:** Structured data, such as user profiles, past design projects, and saved sessions.
 - **Reason:** Relational databases are strong in consistency and well fit for managing structured, user-centric data.
- **Cloud Storage** usages include AWS S3, Google Cloud Storage.
 - **Stores** large files like 3D models, textures, and environment maps.
 - **Reason:** Cloud storage is highly scalable and can hold the big datasets required by high-resolution architectural models.
- **Local Storage** (temporary cache):
 - **Usage:** Temporary caching stores 3D models, textures, and scanned environments for quick access within a session.
 - **Why:** In essence, local caching improves the performance by allowing the system to fetch generally accessed files at incredible speed.

Data Flow and Interaction

The 3D model is loaded from the database or cloud storage and rendered in real-time. Whatever changes are made by the users are immediately applied and stored in the database. The on-set environmental setup detects the spatial data through environment scanning, which ARCore further processes to align the virtual models. AI suggestions populate concerning the input provided by the user and connect with customization option settings to implement instant design changes.

With this in mind, data structures are bound in the system to best handle all the complexities of real-time AR rendering, user customization, spatial mapping, and AI-driven recommendations to make sure the experience by the user is responsive and engaging.